

Sha Yi

She/Her

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Research Interests

Using machine learning to design, control, and simulate novel robotic systems. Data-driven approaches that jointly optimize hardware, fabrication processes, and control policies to embed intelligence in physical structures.

Current Employment

Postdoc Scholar

Sep 2024 - now

UC SAN DIEGO

Work with Xiaolong Wang & Mike Tolley on machine learning for robot design.

Education

Carnegie Mellon University

2024

PHD IN ROBOTICS

Advisor: Katia Sycara, Zeynep Temel

Carnegie Mellon University

2019

MS IN ROBOTICS

Advisor: Katia Sycara

The Hong Kong Polytechnic University

2017

BENG IN ELECTRONIC AND INFORMATION ENGINEERING

Publications

TacO: Benchmarking Tactile Sensors for Object Manipulation

Anya Zorin, Zilin Si, Myungsun Park, Junsung Park, Alexiy Buynitsky, Sachin Bhadang, Taejun Park, Sohee John Yoon, Yong-Lae Park, Oliver Kroemer, Zeynep Temel, Michael T. Tolley, **Sha Yi**, Xiaolong Wang
under review, 2026

Contact-Aware Neural Dynamics

Changwei Jing, Jai Krishna Bandi, Jianglong Ye, Rocky Duan, Pieter Abbeel, Xiaolong Wang, **Sha Yi**
Computer Vision and Pattern Recognition (CVPR), 2026

Data-driven Modeling and Feedforward Control of Millimeter-Scale Vine Robots

Sukjun Kim, **Sha Yi**, Xiaolong Wang, Michael T. Tolley, Tania K. Morimoto
International Conference on Soft Robotics (RoboSoft), 2026

Cross-Embodied Co-Design for Dexterous Hands

Kehlani Fay, Darin Anthony Djapri, Anya Zorin, James Clinton, Ali El Lahib, Michael T. Tolley, Hao Su, **Sha Yi**, Xiaolong Wang
International Conference on Learning Representations (ICLR), 2026

Learning to Design Soft Hands using Reward Models

Xueqian Bai, Nicklas Hansen, Adabhav Singh, Michael T. Tolley, Yan Duan, Pieter Abbeel, Xiaolong Wang, **Sha Yi**

International Conference on Robotics and Automation (ICRA), 2026

Co-Design of Soft Gripper with Neural Physics

Sha Yi*, Xueqian Bai*, Adabhav Singh, Jianglong Ye, Michael T Tolley, Xiaolong Wang

Conference on Robot Learning (CoRL), 2025

Humanoid Policy ~ Human Policy

Ri-Zhao Qiu, Shiqi Yang, Xuxin Cheng, Chaitanya Chawla, Jialong Li, Tairan He, Ge Yan, David J Yoon, Ryan Hoque, Lars Paulsen, Ge Yang, Jian Zhang, **Sha Yi**, Guanya Shi, Xiaolong Wang

Conference on Robot Learning (CoRL), 2025

Reconfigurable Robot Swarms for Terrain Traversal with Passive Coupling Mechanisms

Sha Yi, Shashwat Singh, Allison Seo, Ryan St. Pierre, Katia Sycara, Zeynep Temel

Autonomous Robots, 49(3), 1-18, 2025

Hardware Optimization for In-Hand Rotation

Kehlani Fay, **Sha Yi**, Michael T Tolley, Xiaolong Wang, Hao Su

Workshop on Robot Hardware-Aware Intelligence (RSS), 2025

Mobile-TeleVision: Predictive Motion Priors for Humanoid Whole-Body Control

Chenhao Lu*, Xuxin Cheng*, Jialong Li*, Shiqi Yang, Mazeyu Ji, Chengjing Yuan, Ge Yang, **Sha Yi**, Xiaolong Wang

International Conference on Robotics and Automation (ICRA), 2025

ACE: A Cross-platform Visual-Exoskeletons for Low-Cost Dexterous Teleoperation

Shiqi Yang, Minghuan Liu, Yuzhe Qin, Runyu Ding, Jialong Li, Xuxin Cheng, Ruihan Yang, **Sha Yi**, Xiaolong Wang

Conference on Robot Learning (CoRL), 2024

Decentralized Multi-Robot Line-of-Sight Connectivity Maintenance under Uncertainty

Yupeng Yang, Yiwei Lyu, Yanze Zhang, **Sha Yi** and Wenhao Luo

Robotics: Science and Systems (RSS), 2024

Enhancing Heterogeneous Swarm Locomotion Through Simple 1-DOF Arm Mechanisms

James Clinton, **Sha Yi**, and Zeynep Temel

Distributed Autonomous Robotic Systems (DARS), 2024

*Workshop in Tensegrity Robots, IROS, 2023, **Best Demo Award***

Decentralized Model Predictive Control for Constrained Multi-Robot System

Allison J. Seo, **Sha Yi**, and Katia Sycara

Workshop in Advances in Multi-Agent Learning, IROS, 2023

Reconfigurable Robot Control Using Flexible Coupling Mechanisms

Sha Yi, Katia Sycara, and Zeynep Temel

Robotics: Science and Systems (RSS), 2023

Configuration Control for Physical Coupling of Heterogeneous Robot Swarms

Sha Yi, Zeynep Temel, and Katia Sycara

International Conference on Robotics and Automation (ICRA), 2022

PuzzleBots: Physical Coupling of Robot Swarms

Sha Yi, Zeynep Temel, and Katia Sycara

IEEE International Conference on Robotics and Automation (ICRA), 2021

Distributed Topology Correction for Flexible Connectivity Maintenance in Multi-Robot Systems

Sha Yi, Wenhao Luo, and Katia Sycara

IEEE International Conference on Robotics and Automation (ICRA), 2021

Multi-agent Deception in Attack-Defense Stochastic Game

Xueting Li, **Sha Yi**, and Katia Sycara

International Symposium on Distributed Autonomous Robotic Systems (DARS), 2021

Adaptive Informative Sampling with Environment Partitioning for Heterogeneous Multi-Robot Systems

Yunfei Shi, Ning Wang, Jianmin Zheng, Yang Zhang, **Sha Yi**, Wenhao Luo, and Katia Sycara

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020

Behavior Mixing with Minimum Global and Subgroup Connectivity Maintenance for Large-Scale Multi-Robot Systems

Wenhao Luo, **Sha Yi**, and Katia Sycara

IEEE International Conference on Robotics and Automation (ICRA), 2020

Indoor Pursuit-Evasion with Hybrid Hierarchical Partially Observable Markov Decision Processes for Multi-Robot Systems

Sha Yi, Changjoo Nam, Katia Sycara

International Symposium on Distributed Autonomous Robotic Systems (DARS), 2018

Previous Employments

Amazon Robotics

APPLIED SCIENTIST INTERNSHIP

with *Dr. Andreas Kolling* on Multi-robot planning and control.

North Reading, MA

Jun 2022 - Aug. 2022

Google Summer of Code

DEVELOPER

with *Prof. Kei Okada* on collision avoidance for manipulation and humanoid, JSK Robotics Lab of University of Tokyo.

Virtual

May 2017 - Aug. 2017

HAI Robotics

ROBOTICS INTERNSHIP

Implemented path planning algorithm for warehouse automation.

Shenzhen, China

Mar. 2016 - May 2016

Microsoft

SOFTWARE ENGINEER

Cloud and Enterprise division, Platform and Tools group.

Beijing, China

Jul. 2015 - Dec. 2015

Honors & Awards

MIT EECS Rising Star	2025
CMU Presidential Fellowship	2021
HKSAR Government Scholarship	2014, 2015, 2016

Academic Services

Reviewer	Conference: ICRA, IROS, WAFR, ACC, RoboSoft, RSS, CoRL; Journal: TRO, RAL, AURO
Workshop Organizer	ICRA 2024: Unconventional Robots: Universal Lessons for Designing Unique Systems RSS 2025: Brain and Brawn – Robot Hardware-Aware Intelligence CoRL 2025: Human to Robot: Sensorizing, Modeling, and Learning from Humans

Teaching Experiences

Math Fundamentals for Robotics	Fall 2019, CMU
Kinematics, Dynamics, and Control	Spring 2020, CMU
Soft Robots	Fall 2024, 2025, UCSD
Machine Learning for Mechanical Engineers	Spring 2026, UCSD

Diversity & Outreach Services

Robotics Institute Summer Scholars (RISS)	2019, 2020, 2021, 2023
Served on the admission committee and reviewed applications. Mentored undergraduate students for three-month research projects.	
Women@SCS/SCS4ALL Mentoring Program	2020, 2021, 2022
Mentored undergraduate students from underrepresented backgrounds. Introduced students to research and helped them shape their career paths.	
SCS Graduate Application Support Program (GASP)	2020, 2021, 2022, 2023
Helped underrepresented students from outside of CMU for graduate school applications. Provided advice on resume and personal statements.	

Talks

Learning to Design Robots	2026
Invited talk, University of Notre Dame Invited talk, Virginia Tech Invited talk, Purdue Invited talk, NYU Invited talk, UChicago	
Learning to Design Soft Robots	2025
Invited talk, Global Soft Robot Day, IEEE RAS Technical Committee Guest Lecture, Soft Robot, Fall 2025 UCSD Invited talk, Feed the Intellect Robotics Seminar, UCSD Invited talk, Robotics Discussion Group, Duke	

Reconfigurable Robot Control Using Flexible Coupling Mechanisms

2024

Guest Lecture, Soft Robot, Fall 2024 UCSD

Improving Robot Capabilities Through Reconfigurability

2024

Invited talk, University of British Columbia

Invited talk, UCSD (Host: Xiaolong Wang & Mike Tolley)

Invited talk, REALM Lab, MIT (Host: Chuchu Fan)

Invited talk, Sung Robotics Lab, UPenn (Host: Cynthia Sung)

Physical Coupling in Robot Swarms

2023

Guest lecturer, Insects and Robots, Fall 2023 CMU

Workshop on Tensegrity Robotics, IROS

Filling in the Gaps: Physical Coupling for Reconfigurable Robots

2022

Workshop on Modular Self-Reconfigurable Robots, ICRA